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Project Title:	Hypobaric Hypoxia in Space Data Mining		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Flight Program:	Shuttle		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wessel 3rd, James M.S. (Wyle Laboratories, Inc.)		
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	Extravehicular activity (EVA) is a vital component of human space exploration. The pressure differential between the internal cabin of a spacecraft and a space suit pressurized to 4.3 psia introduces the risk of decompression sickness (DCS). The risk of DCS is currently reduced with lengthy denitrogenation protocols. To reduce both the denitrogenation time and the risk of DCS during EVA, the option of an intermediate habitat pressure of 8.2 psia with 34% oxygen (herein referred to as 8.2/34) has been proposed for NASA's exploration missions. However, the proposed atmosphere creates an inspired oxygen partial pressure (PIO2) of 128 mm Hg, which is equivalent to breathing air at approximately 4,000 ft (1220 m) altitude. Coincidentally, one of the denitrogenation protocols used during the Shuttle Program was a staged protocol that exposed astronauts to 10.2 psia and 26.5% oxygen (10.2/26.5). The Shuttle-staged protocol resulted in a mildly hypoxic PIO2 of 127 mm Hg, almost identical to that of the proposed 8.2/34 environment. Terrestrial data suggests that this hypoxic dose is acceptable for unimpeded mission operations; however, the interaction between		

Task Description:	adaptation to microgravity, the spaceflight environment, and mild hypobaric hypoxia is uncertain. Aims: 1. Determine accurate environmental exposure doses and durations for crewmembers during the shuttle program including hypoxia and elevated CO2. 2. Identify and evaluate any available flight data from the Shuttle program that may aid in clarifying the physiologic and health effects of exposure to mild hypobaric hypoxia in microgravity. 3. Review terrestrial hypoxia literature with respect to very mild hypoxia at altitudes of 6000 feet or less.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. In order to determine exact times of flight days and for the exact times in which the shuttle cabin is depressed to 10.2 psia (thus creating the hypoxic condition), a researcher needs to have access to the Archive Data Retrieval (ADRIFT) subprogram inside of the Java Mission Evaluation Workstation System (JMEWS), which is only available on site at Johnson Space Center, and requires proper clearance.
Research Impact/Earth Benefits:	Expected results of this effort will inform the acceptability of using the exploration enhancing capability of the NASA Exploration Atmosphere of 8.2 psia and 34% oxygen. This will provide significant improvement to extravehicular activity (EVA) capabilities and reduce consumables associated with EVA.
	SUPPLEMENTAL FINAL REPORTING IN OCTOBER 2016 Spacecraft, planetary rovers, and habitats for NASA's future human exploration missions requiring frequent extravehicular activity are expected to use an atmospheric composition of 34% oxygen (O2) and 66% nitrogen (N2) at 56.5 kPa (8.2 psia) to reduce time and consumables required for denitrogenation without exceeding flammability constraints. This atmosphere will provide a mild level of hypoxia (PIO2 = 128 mmHg) approximately equivalent to that experienced during portions of 41 Space Shuttle missions that used a "Staged" prebreathe protocol starting from an atmosphere of 26.5% O2 and 73.5% N2 at 70.3 kPa (10.2 psia) (PIO2 = 127 mmHg). METHODS: Retrospective environmental data were examined to determine shuttle mission time at 70.3 kPa (10.2 psia) and ppO2 and ppCO2 values for missions that performed a Staged prebreathe protocol. Private medical conferences and post-flight debriefs were examined and data related to 14 possible hypoxia symptoms were compared with equivalent data collected during normoxic Space Shuttle operations at 101.3 kPa (14.7 psia) using logistic regression. RESULTS: Over the Space Shuttle program, there were 134.1 days (788.8 person-days adjusted for crew size) under the 70.3 kPa (10.2 psia) Staged condition (average 3.17 ± 2.2 days per mission, maximum 8.1 days) and 258.81 days at 101.3kPa (14.7 psia) and 2192.95 total person-days of normoxic operations. An average of 4.31 potentially hypoxia-related symptoms per mission day was documented under Staged conditions compared with 4.08 per mission day during normoxic operations. Logistic regression analysis found none of the hypoxia signs and symptoms to be significantly associated with the 70.3 kPa (10.2 psia) condition versus the 14.7 psia condition. DISCUSSION: Terrestrial experience indicates that chronic exposure to a PIO2 of 876 kPa (137 mmHg) (1219 m altitude) is well-tolerated by healthy humans. Shuttle experience indicates that similar levels of hypoxia in space at a pressure of 70.3 kPa (10.2 psia) resulted in no increased reporting of possible hypoxia related symptoms, and no obvious crewmember performance issues related to completion of mission objectives. However, these results may not be representative of how chronic mild hypoxia interacts with physiologic changes due to microgravity over long duration missions. A manuscript has been submitted to Aerospace Medicine and Human Performance with title "No Evidence of Clinical Symptoms due to Mild Hypobaric Hypoxia Exposure in Microgravity." ANNUAL REPORTING AS OF AUGUST 2015 Introduction – Recently, NASA has decided to establish a defined Exploration Atmosphere for all future manned discovery of the Solar System [1,2]. Originally it was defined as 8.0 psi and 32% Oxygen (remainder Nitrogen), but has since been improved, from a hypoxia standpoint, to 8.2 psi 34% O2 (hereafter, 8.2/34). This exploration atmosphere was designed with a goal of maximizing 'boots on the surface' EVA time vs. prebreathe denitrogenation time for the prevention of Decompression Sickness (DCS), against the threat of increased oxygen flammability. The net effect of the combination of 8.2/34 is an inspired oxygen partial pressure (PIO2) of 128 mmHg. This corresponds almost exactly to the 10.2 (26.5% O2) staged Shuttle condition with a PIO2 of 127 mmHg O2. On the basis of this reasoning, it was thought that Shuttle missions that featured the 10.2 condition would serve as a historical spaceflight comparison for the concerns of mild hypobaric hypoxia (HH) found in the future Exploration Atmosphere. 40 Shuttle missions used the 10.2 Staged protocol, and form the basis for the 10.2 data mining effort. The underlying goal is to examine the human health and physiologic results from those distinct Shuttle missions and use them as a data source to help predict any consequences of mild HH on astronauts' physiology for future manned exploration. Methods – NASA's Lifetime Surveillance of Astronaut Health (LSAH) The Lifetime Surveillance of Astronaut Health (LSAH) is a proactive occupational surveillance program for the astronaut corps to screen and monitor astronauts for occupational related injury or disease. This active archive that provides information and data from 1961 (Mercury Project) through current flight and flight analog studies (International Space Station, Shuttle, bed rest studies, etc.) involving human, plant, and animal subjects. Through Private Medical Conferences and Postflight Medical Debrief documents for crewmembers exposed to the 10.2/26.5 condition, the LSAH database provided astronaut reports of symptomatic complaints such as shortness of breath, headache, lethargy/malaise, sleep disturbances, nausea/vomiting, dizziness, confusion/disorientation, appetite changes, etc., between 10.2 depress shuttle missions and normal 14.7 psia shuttle missions.

Task Progress:

Previous data mining efforts evaluated 112 LSDA (Life Science Data Archive) studies (Detailed Supplementary Objectives - DSO) that were performed during one or more missions that were exposed to the staged condition. These studies were prioritized and evaluated based on physiologic relevance and number of available subjects. The 'Sleep-Wake Actigraphy and Light Exposure During Spaceflight' study (DSO 634) was identified as having potentially relevant data of a sufficient sample size.

From a shuttle cabin pressure/atmospheric viewpoint, the EVA physiology group has collected distinct times of shuttle cabin depress/repress cycles, and oxygen and carbon dioxide levels for all shuttle missions via the Archive Data Retrieval (ADRIFT) subprogram inside of the Java Mission Evaluation Workstation System (JMEWS) and the Operational Prebreathe Database. This crewmember exposure history to the Shuttle-staged protocol will be correlated with medical and physiologic data available in the Lifetime Surveillance of Astronaut Health (LSAH) database (outlined above).

Additionally, the study team will examine available literature of terrestrial studies in which altitude exposures are 4000ft or less for direct comparison to the data mined spaceflight evidence outlined above.

Results:

LSAH analysis – Of the 135 Shuttle missions, 126 were included in the analysis. Missions were not included due to unavailable medical debriefs (8) or the use of a different depress protocol other than the 10.2 protocol (1). 40 missions used the 10.2 depress protocol while 86 used the 14.7 protocol. 250 distinct astronauts were in the data resulting in 521 person missions.

Logistic regression was used to determine the relationship between reporting the presence of space adaptation symptoms and the amount of time spent in the depressed, hypoxic environment. A separate regression was run for each of the 14 symptoms. For each regression equation, the odds of a reported symptom was the outcome and the amount of time spent at depress was the predictor. The covariates gender, age at launch, EVA participation, and docking with either International Space Station (ISS) or Mir were included in the analysis. The regression also adjusted for astronauts who flew on multiple missions. The regressions were conducted using the total number of days with a symptom reported (as the number of events) over the number of flight days (s the number of trials). This method was used due to the inability to determine whether a symptom occurred during a time of depress or even on the day in which a depress occurred.

Although 14 different symptoms were in the data, only 10 regression equations could be produced due to a low reporting rate of four of the symptoms (drowsiness, irritability, loss of initiative or motivation, and other). For all 10 equations, there was not a significant effect of time spent in depress on the odds of reporting a symptoms. The coefficient of time in depress for each equation had a p-value greater than .005 (Bonferroni adjusted a). From this, we can find no statistical evidence that HH was predictive of these symptoms.

DSO 634 findings – There are six astronauts in the sleep database who were exposed to the decreased atmospheric pressure with slightly hypoxic conditions on 10.2 Staged shuttle missions. Sleep during these missions will be compared to sleep on other shuttle missions with different atmospheric pressure and gas constituency. Although a cohort of only 6 astronauts may not provide power to realize significant differences in sleep associated with hypoxia, the EVA Physiology Laboratory feels that this exploratory research is important and will contribute preliminary data to guide the design and implementation of the future Exploration Atmosphere.

Another stealthy environmental factor that could affect sleep is hypoxia. Appropriate oxygenation is achieved and maintained by various combinations of percentage of oxygen (O2) and atmospheric pressure. The 10.2 depress missions represented a slightly hypoxic condition that may have impacted the astronauts' respective sleep.

The analysis of DSO 634 has been determined to be feasible but funding, Institutional Review Board (IRB) approval and data share agreements still need to be worked out.

Shuttle Cabin pressure/atmosphere findings –

According to flight data harvested from JMEWS, there were 132.8 days (782.2 man-days, when adjusted for crew size) under the 10.2 staged condition. Length of exposure to this environment varied by mission. The majority of missions had 10.2 exposure durations under 72 hrs. The overall average time was about 3.22 days and the longest being just over 8 days. Notably, 7 shuttle missions had interrupted 10.2 psia staged conditions, where the shuttle depressed, conducted one or more EVAs, then repressed to 14.7 psia. After a brief interval at 14.7 psia, the cabin was again depressed to 10.2 psia for one or more additional EVAs. For the atmospheric cabin gases, ppO2 averaged 142.2 ± 3.4 mmHg while depressed to 10.2, and 163.8 ± 6.2 mmHg at 14.7 psia before and after depress across all staged missions. No lower than expected O2 events (<2.35 ppO2) were seen in flight data. The ppCO2 varied among the 10.2 shuttle missions, the average was 1.97 ± 0.6 mmHg while at 10.2 psia and 2.14 ± 0.8 mmHg at 14.7 psia. 29 shuttle missions had peak ppCO2 over 4 mmHg with 6 missions above 7.6 mmHg. 5.5% of all ppCO2 values were above 4 mmHg, and 0.07% were found to be above 7.6 mmHg. The maximum recorded ppCO2 on any 10.2 depress mission was 11.02 mmHg.

Literature review of less than 4000 ft altitude –

Simply stated, terrestrial hypobaric hypoxia concerns at sub 4000 ft altitude are almost non-existent except for the pulmonary challenged or the aged and infirm. The literature reflects this pauperacy. As a result, the examined range or our informational search was expanded to include findings up to 10,000 ft altitude. Here we realized some results that could potentially be noteworthy for the Exploration Atmosphere. First, we'll briefly state the positives. Federal regulations allow commercial aircraft to have cabin pressures reach the 8,000 ft mark, but most are lower. There are remarkably few findings of physiologic compromise from this atmospheric pressure, unless the passenger was health impaired already. That represents a vast everyday traveling population. There is a large population of people that live daily at altitudes at or in excess of 4,000 ft. Again, if this was physiologically challenging beyond diseased or aged, people wouldn't choose to live there. Now, we'll briefly recap the mildly negative findings. A several facets of vision (dark adaptation, contrast sensitivity, color sensitivity, mesopic and scotopic vision, and visual response) are impacted at the 8,000-10,000 ft range of altitude. Sleep may be impaired (e.g., arousals and awakenings), though it typically returns to normal after adaptation. The comprehensive literature review is still underway.

Summary –

As our work continues, even null findings will have value. If atmosphere related health issues (e.g., headaches and sleep

	disruption at 10.2/26.5) were not recorded for the shuttle missions with an inspired oxygen partial pressure (PIO2) of 127 mmHg, then we should expect limited or no operational impact for the 8.2/34 exploration atmosphere with a nearly equivalent PIO2 of 128 mmHg. Literature cited: [1] Campbell and Henninger, “NASA Exploration Atmospheres Working Group Recommendations for Exploration Spacecraft Internal Atmospheres: The Final Report of the NASA Exploration Atmospheres Working Group” NASA/TP-2010-216134, 2010. [2] Norcross et al., “Effects of the 8 psia / 32% O2 Atmosphere on the Human in the Spaceflight Environment” NASA/TM-2013-217377, 2013. LSDA/LSAH Website: https:// NOTE the following presentations are related to this work, while it was a pilot study prior to this project: Wessel JH 3rd, Norcross JR. "Examination of 10.2 PSIA Shuttle Missions as a Precursor to the Future Exploration Atmosphere." Presented at the 2014 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/ (Abstract #3099) Thompson MS, Norcross JR, Klein JS, Wessel JH 3rd. "Evaluation of Crew Exposure to Mild Hypobaric Hypoxia in a Microgravity Environment." Presented at the 2013 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 12-14, 2013. (Workshop website and program are currently not publicly available)
Bibliography Type:	Description: (Last Updated: 01/08/2026)
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